

必记忆内容:

(0) Rendering pipeline.

- (1) world coordinate transformation
- (2) perspective transformation
- (3) back-face removal
- (4) clipping
- (5) rasterization
- (6) hidden surface removal
- (7) shading

1.Object Modeling

(1) Representations of 3D object

Point clouds;
Polygonal meshes;
Subdivision surfaces;
Implicit surfaces;
Parametric surfaces;;
Voxel;
Constructive solid geometry;
Fractals;

(2) 2D Drawing, Drawing a Thin Line

- Answer: 1) Set x to x_1 and y to y_1 , and shade this pixel;
2) Increase x by 1, and correspondingly $y = y + m$
3) Compute D_1 – distance of (x, y) from the center of the upper pixel;
4) Compute D_2 – distance of (x, y) from the center of the lower pixel;
5) If $D_1 < D_2$ (i.e., the line is closer to the upper pixel), shade the upper pixel;
otherwise, shade the lower pixel.
6) if the endpoint (x_2, y_2) is not achieved, turn to 2); otherwise, stop.
(3) Region Filling (区域填充): 可以用 Flood-fill algorithm (泛洪填充算法)

(4) Polygon Meshes (多边形网格): 三元组 (V, E, F)

V 是顶点集合 (set of vertices)。 E 是边集合 (set of edges)。 F 是面集合 (set of faces)

凸多面体, 无洞: $V - E + F = 2$

亏格 g , 边界数 b : $V - E + F = 2 - 2g - b$

- (5) parametric equation for a 3D sphere of radius r centered at origin.

$$\begin{cases} x = r \sin \phi \cos \theta \\ y = r \sin \phi \sin \theta \\ z = r \cos \phi \end{cases}$$

$\theta \in [0, 2\pi)$

$\phi \in [0, \pi]$

2. Projection and Clipping

(1) perspective projection

Provides a sense of depth (objects appear larger when closer and smaller when farther away).

Lines not parallel to the projection plane undergo foreshortening (perspective distortion).

Such lines converge toward a vanishing point.

(2) cop 移远

1. Foreshortening (透视缩短): This reduces foreshortening, making the object appear "flatter."

2. Vanishing Points: The vanishing points move further apart toward infinity.

3. Depth Sensation: The "depth sensation" is reduced;

(3) 为何先透视变换再裁剪

Before: clipping planes are arbitrary in 3D $\rightarrow$ complex line-plane intersection

After transform: planes become axis-aligned

$x = \pm A, y = \pm B, z \in [d, C]$

Clipping simplifies to range checks instead of 3D intersection

$\rightarrow$ reduces complexity

(4) Perspective Transformation 的优点

Core: frustum $\rightarrow$ axis-aligned box + preserve z

1. Simplifies Back-face Removal

2. Simplifies Clipping

3. Enables Depth Processing (Z-buffer)

3.shading

(1) light type

环境光 (Ambient light):

点光源 (Point light source)

聚光灯 (Spotlight / searchlight)

面光源 (Area light source)

(2) Phong 光照模型 (Phong Illumination Model)

I_c : 环境光的强度; I_l : 入射光强度

环境光反射 (Ambient reflection) : $I_a = k_a \cdot I_c$

漫反射 (Diffuse reflection) : $I_d = k_d \cdot I_L \cdot \cos \theta = k_d \cdot I_L \cdot \max(0, N \cdot L)$

镜面反射 (Specular reflection): $I_s = k_s \cdot I_L \cdot \cos^n \phi = k_s \cdot I_L \cdot \max(0, V \cdot R)^n$

$R = L - 2(N \cdot L)N$

(3) Why Gouraud shading misses specular highlights:

Interpolates color, not normal

→ The specular term $(V \cdot R)^n$ is non-linear

→ For large n , it produces a sharp peak (highlight)

→ Hence, it cannot be accurately represented by linear interpolation

4. Aliasing and anti- Aliasing

(1) why Aliasing

Point sampling is a coarse approximation

Aliasing is caused by undersampling of high-frequency signals

(2) anti-aliasing

Supersampling (超采样): easy to realise

only reduce but not eliminate it, still point sampling

considerable amount of memory and processing time

Accumulation buffer (累积缓冲区): address the considerable of memory usage issue (save memory), realize Motion Blur and depth of field

Stochastic sampling (随机采样)

Introduce random noise

Catmull's algorithm (Catmull 算法)

Computationally very expensive

The A-buffer method (A-buffer 方法)

(3) what is Rasterization?

The process of taking a primitive and figuring out which pixels it covers

5. Ray Tracing

(1) The process

Cast ray from cop → find intersection → Cast shadow rays and compute Phong illumination (direct lighting) → trace reflection/refraction → combine colors: $C_{\text{pixel}} = C_{\text{direct}} + iC_{\text{reflection}} + tC_{\text{refraction}}$

(2) The advantages and disadvantages

Adv: Accurate shadows; Realistic reflection and refraction; Good global illumination effects

Disadv: computationally expensive; Not suitable for real-time rendering; Weak at indirect diffuse reflection; View-dependent rendering

6. Radiosity

(1) The advantages and disadvantages

Adv: View-Independent; can handle diffuse reflection

Disadv: computationally expensive; Weak at specular reflection

7. Real-Time Rendering

(1) Shadow Volume vs Shadow Map 对比

Shadow volume: accurate, non-aliasing; add complexity

Shadow map: easy; aliasing

(2) Efficient Communication 为什么重要

efficient communication ensure that GPU doesn't waste time moving data and waiting.